

- -OH and CH_3 is electron donating group where -OH is stronger.
- $-NO_2$ is an electron withdrawing group.
- Due to srtrong donating tendency of -OH, attack of electrophile ($-NO_2^+$) will be governed by -OH and $-NO_2^+$ will attack at para position w.r.t -OH.

Therefore, **Option(3) is correct.**

Q. 11 If you spill a chemical toilet cleaning liquid on your hand, your first aid would be:

Option 1:
aqueous NH_3

Option 2:
gaseous $NaHCO_3$

Option 3:
aqueous NaOH

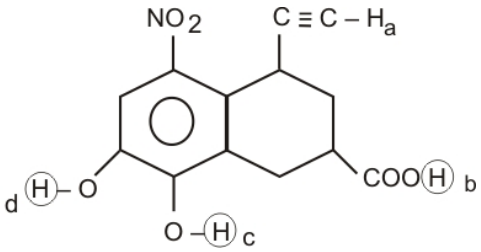
Option 4:
Vinegar

Correct Answer:
gaseous $NaHCO_3$

Solution:
Chemical toilet cleaning liquid contains acid and hence the first aid would be a weak base like $NaHCO_3$.

Therefore, **Option(2) is correct.**

Q. 12 Arrange the following labeled hydrogens in decreasing order of acidity:



Option 1:
c > b > a > d

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Q. 20 Let α and β be the roots of $x^2 - 3x + p = 0$ and $\gamma \ and \ \delta$ be the roots of $x^2 - 6x + q = 0$. If $\alpha, \beta, \gamma, \delta$ forms a geometric progression. Then the ratio $(2q + p) : (2q - p)$ is :

Option 1:
3:1

Option 2:
9:7

Option 3:
5:3

Option 4:
33:31

Correct Answer:
9:7

Solution:
The general term or n[th] term of a geometric progression is $a_n = ar^{n-1}$

Now,

$x^2 - 3x + p = 0$
roots are α & β
$\alpha + \beta = 3$
$\alpha\beta = p$

$x^2 - 6x + q = 0$
roots are γ & δ
$\gamma + \delta = 6$
$\gamma\delta = q$

α, β, γ & δ are in GP
Let, $\alpha = a, \ \beta = ar, \ \gamma = ar^2$ and $\delta = ar^3$

$a + ar = 3 \qquad \ldots(1)$
$ar^2 + ar^3 = 6 \qquad \ldots(2)$
$(2) \div (1)$
$r^2 = 2$

So, $\dfrac{2q+p}{2q-p} = \dfrac{2r^5 + r}{2r^5 - r} = \dfrac{2r^4 + 1}{2r^4 - 1} = \dfrac{9}{7}$

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Q. 21 Suppose differntial function f(x) satisfies the identity

$f(x+y) = f(x) + f(y) + xy^2 + x^2y$, for all real x any y. If $\lim_{x\to 0} \frac{(x)}{x} = 1$ then f'(3) is equal to _____

Correct Answer:
10

Solution:

$$\text{Since, } \lim_{x\to 0} \frac{f(x)}{x} \text{ exist } \Rightarrow f(0) = 0$$

$$\text{Now, } f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h\to 0} \frac{f(h) + xh^2 + x^2h}{h} (\text{ take } y = h)$$

$$= \lim_{h\to 0} \frac{f(h)}{h} + \lim_{h\to 0}(xh) + x^2$$

$$\Rightarrow f'(x) = 1 + 0 + x^2$$

$$\Rightarrow f'(3) = 10$$

Q. 22 If the system of equations

$x - 2y + 3z = 0$

$2x + y + z = b$

$x - 7y + az = 24$, has infinitely many solutions, then a-b is equal to_____

Correct Answer:
5

Solution:

$$\text{D} = \begin{vmatrix} 1 & -2 & 3 \\ 2 & 1 & 1 \\ 1 & -7 & 8 \end{vmatrix} = 0 \Rightarrow \text{a} = 8$$

$$\text{also, } \text{D}_1 = \begin{vmatrix} 9 & -2 & 3 \\ \text{b} & 1 & 1 \\ 24 & -7 & 8 \end{vmatrix} = 0 \Rightarrow \text{b} = 3$$

$$\text{hence, a} - \text{b} = 8 - 3 = 5$$

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Option 3:
a - (iii), b - (i), c - (i), d - (iv)

Option 4:
a - (iv) , b - (ii), c - (i), d - (iii)

Correct Answer:
a - (iv) , b - (ii), c - (i), d - (iii)

Solution:
Correct Answer: 4) a - (iv) , b - (ii), c - (i), d - (iii)

The radiation ad corresponding wavelengths are given below

Microwave-$10^{-3}m$

AM Radio waves-$100m$

Gamma rays-$10^{-15}m$

X-rays- $10^{-10}m$

Q. 19 A parallel plate capacitor has a plate of length l, width w, & separation of plates is d. If it is connected to a battery of emf V, A dielectric slab of the same thickness d & of dielectric constant K = 4 is being inserted between the plates of the capacitor. At what length of the slab inside states, will the energy stored in the capacitor be two times its initial energy stored?

Option 1:
$\frac{2l}{3}$

Option 2:
$\frac{l}{2}$

Option 3:
$\frac{l}{3}$

Option 4:
$\frac{l}{4}$

Correct Answer:
$\frac{l}{3}$

Solution:

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Correct Answer: 3) $\frac{l}{3}$

Let the dielectric insert up to length x

For dielectric inserted part

$$C_1 = \frac{K\epsilon_0 * w * x}{d}$$

and for the remaining part

$$C_2 = \frac{\epsilon_0 * w * (l - x)}{d}$$

Now,

$$C_{eq} = C_1 + C_2 = \frac{\epsilon_0 * w}{d} * (l + x(k - 1))$$

$$U_{final} = \frac{1}{2}C_{eq}V^2 = \frac{\epsilon_0 * w * V^2}{2d} * (l + x(k - 1))$$

$$U_{initial} = \frac{1}{2}CV^2 = \frac{\epsilon_0 * w * l * V^2}{2d}$$

But, $U_{fial} = 2U_{initial}$

On solving we get $x = \frac{l}{3}$

Q. 20 A ring is hung on a nail. It can oscillate without slipping or sliding (i) in its plane with a time period T_1 and (ii) back and forth in a direction perpendicular to its plane with a period T_2 The ratio $\frac{T_1}{T_2}$ will be

Option 1:
$\frac{\sqrt{2}}{3}$

Option 2:
$\frac{2}{3}$

Option 3:
$\frac{2}{\sqrt{3}}$

Option 4:
$\frac{3}{\sqrt{2}}$

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